A COMBINATION ATTACK

A.1 Overview

While gradient-based methods like GASLITE [3] can manipulate retrieval rankings, they require "white-box" access to model internals. The Combination Attack bridges this gap as a black-box methodology, hijacking the retrieval process using only tokenizer and embedding outputs. Crucially, this approach achieves performance parity with white-box states-of-the-art, consistently reaching perfect top-1 success rates across diverse open- and closed-source models.

A.2 Methodology

The Combination Attack is a multi-stage optimizer that appends an adversarial "trigger" suffix to a malicious passage ("info"), making it semantically indistinguishable from a target passage in vector space. It uses a hybrid approach to maximize similarity through three distinct strategies:

Phase 0: Hot-Start. Before iterative optimization, the attack employs a "Hot-Start" initialization to anchor the adversarial passage in the target's semantic domain. An auxiliary Large Language Model (LLM) is queried to generate 10-20 tokens contextually relevant to the target query. This provides a good initial state, significantly reducing the number of steps required in later phases by starting the search in a high-similarity region of the embedding space.

Phase 1: Random Search. Starting from the hot-start seed, the algorithm generates a pool of random candidate tokens from the model's vocabulary. Each candidate is evaluated based on the resulting cosine similarity between the adversarial and target passages. The highest-scoring token is permanently appended, grounding the sequence in a promising semantic region.

Phase 2: Square-like Attack. The final stage refines the sequence using a block-based optimization adapted from the Square Attack [1]. Instead of pixel patches, this adaptation operates on contiguous token segments. The algorithm iteratively mutates random segments; replacements are accepted only if they increase the cosine similarity against the target embedding. By decreasing the perturbation window over time, the attack transitions from broad exploration to precise semantic alignment.

A.3 Experimental Evaluation

The efficacy of the Combination Attack was evaluated against nine popular open-source embedding-based models and three state-of-the-art closed-source embedding models. The results are averaged over 50 trials; in each trial, a unique random query ("target passage") and a random malicious passage ("info") were chosen.

Open-Source Models. MiniLM [12], aMPNet [9], Arctic [6], E5 [11], GTR-T5 [7], Contriever [5], Contriever-MS [5], ANCE [13], mMPNet [9].

Closed-Source Models. OpenAI's text-embedding-3-small (OAI-Small) [8], OpenAI's text-embedding-3-large (OAI-Large) [8], and Google's gemini-embedding-001 (Gem-Large) [10].

Datasets. MSMARCO [2] for the queries and corpus, and ToxiGen [4] for the malicious "info" passages.

Baselines. We tested **info Only** where the adversarial passage is just the chosen info alone; **stuffing**, where the query was appended to the end of info; and **RASLITE**.

Table 1: Success Rate (appeared@1) Across Embedding Models. This table compares the top-1 retrieval success of our proposed Combination Attack against standard baselines and RASLITE.

	appeared@1			
Model	info Only	stuffing	RASLITE	**Combi.**
MiniLM	0%	10%	**100%**	**100%**
aMPNet	0%	18%	**100%**	**100%**
Arctic	0%	38%	**100%**	**100%**
E5	0%	2%	**100%**	**100%**
GTR-T5	0%	8%	76%	**100%**
Contriever	0%	90%	**100%**	**100%**
Contriever-MS	0%	10%	**100%**	**100%**
ANCE	0%	0%	**100%**	**100%**
mMPNet	0%	30%	**100%**	**100%**
OAI-Small[1]	0%	0%	**100%**	**100%**
OAI-Large[1]				
Gem-Large[1]				

The experimental results displayed in Table 1 unequivocally demonstrate the superior performance of the Combination Attack. Across all models, the proposed methodology outperforms all baselines.

Table 2: Token Count Comparison. This table details the total tokens required for Combination Attack and RASLITE, with reduction calculated as the ratio of RASLITE tokens to Combi. tokens.

	Tokens		
Model	RASLITE	**Combi.**	**Reduction** (×)
MiniLM	1,218,702	**124,343**	9.80
aMPNet	3,019,949	**250,816**	12.04
Arctic	1,360,054	**202,426**	6.72
E5	1,892,159	**242,962**	7.79
GTR-T5	3,394,535	**278,040**	12.21
Contriever	701,259	**73,204**	9.58
Contriever-MS	810,559	**83,794**	9.67
ANCE	1,607,391	**229,577**	7.00
mMPNet	1,570,016	**143,010**	10.98
OAI-Small[1]	2,556,917	**714,794**	3.58
OAI-Large[1]			
Gem-Large[1]			
Average	1,813,154	**234,296**	7.74

Additionally, the Combination Attack manages to perform the optimization at a much lower cost compared to RASLITE, on average (across all models) having a reduction factor of 7.74×.

[1]For the closed-source models, the best passage was estimated by using an index on an open-source model to retrieve the top-50 passages, which were then re-embedded to find the maximum similarity; this approach was used to avoid the large cost of indexing the entire corpus with closed-source APIs.

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